

ENGG2112 PROJECT PROPOSAL: BUSH-FIRE DETECTION

(TUT 02 - GROUP 10)

TEAM:

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OBJECTIVES

Machine learning has the capacity to combat the rising prevalence of bushfires and reduce their impacts. These tools can help in:

1. Bushfire Prediction Modelling: analyzes data collected from the satellite images captured from the LANCE FIRMS operated by NASA's Earth Science Data and Information System (ESDIS). These algorithms can then be used to predict the likelihood of fires by looking at the brightness temperature of the pixels recorded in kelvins captured by the satellite in real time.
2. Bushfire detection and monitoring: Machine learning methods can be used to train the machine to analyze the data collected from the satellite images and detect the presence of fire.

We can considerably lessen the impact of bushfires and safeguard our communities and environment from its destructive effects by utilising the power of machine learning. The goal of this project is to improve bushfire prediction, detection, and reaction by gathering and analysing data using machine learning techniques. With this information, we can create bushfire prevention and impact mitigation strategies that are more potent.

We aim to provide governments and fire brigades with the necessary tools to forecast, monitor, and react to bushfires in a more effective and efficient manner by implementing machine learning algorithms in bushfire prediction modelling and detection. As a result, bushfires may be stopped in their tracks, local flora and fauna may sustain less harm, as well as infrastructure and human lives may also be preserved. We can also greatly lessen the impact of bushfires on the animals residing around that area and reduce environmental effects like carbon and greenhouse gas emissions which further deteriorates the air people breathe and negatively impacts their respiratory systems.

JUSTIFICATION

We chose this subject for our project because we all share a passion for conserving wildlife and natural resources. We wanted to work on this subject and try to help society in any way we could because we have all seen numerous instances of bushfires turning into forest fires that are being reported every year. Studying engineering in general develops and improves our problem-solving skills, and these skills can be harnessed to carry out tasks as we complete our objective.

Aidan, majoring in Civil Engineering, has a proficient understanding of the core idea and the justification of minimising the impacts of bushfires. Gayatri and Samiksha, both majoring in Software Engineering, have a strong background in Python and a passion for machine learning.

IMPACT ON SOCIETY:

Bushfires are a natural, essential and complex part of Australia's environment but their ferocious nature has monumental impacts on society and the land. In 2019-2020 alone the bushfires caused an estimated \$4-5 billion worth of economic loss on the Australian food system. Their wrath also burnt an estimated 24.3 million hectares (60 million acres; 243,000 square kilometers; 94,000 square miles), destroyed over 3,000 buildings (including 2,779 homes) and killed at least 34 people. Machine learning as a tool can analyse bushfires to predict, detect and monitor their behaviors, in doing so can be used to reduce the physical damage endured.

The aim of our project is to use the data gathered by satellites to predict bushfires and monitor their behaviors and patterns. This applied machine learning can significantly impact the society by providing communities with early warning systems that enable them to plan for evacuation and lessen the effects of bushfires on life and property. This will ultimately have a positive economic

influence by reducing the financial burden on firefighting services and enabling more efficient allocation of resources.

METHODS

As to complete our main objective of 'Bushfire Detection' we have used the dataset - [AUSTRALIAN BUSHFIRE SATELLITE](#) (Kaggle, 2019).

DATA-PROCESSING AND DATA CLEANING:

The data that we have selected is originally quite clean and has no null or missing values, therefore, we will only do minor changes to the data that are required like using one-hot encoding to convert the values of the 'instrument' feature into binary representation. We will also remove a few columns if they will not be used much in the methods below.

ATTRIBUTE SELECTION:

- **latitude** : Center of nominal 375 m fire pixel
- **longitude** : Center of nominal 375 m fire pixel
- **brightness** : temperature I-4 - VIIRS I-4 or MODIS 21/22 channel brightness temperature of the fire pixel measured in Kelvin.
- **scan** : Along scan pixel size - the algorithm produces approximately 375 m pixels at nadir. Scan and track reflect actual pixel size.
- **track**: Along Track pixel size - The algorithm produces approximately 375 m pixels at nadir. Scan and track reflect actual pixel size.
- **acq_date**: Date of VIIRS/MODIS acquisition.
- **acq_time**: Time of acquisition/overpass of the satellite (in UTC).
- **satellite**: N= Suomi National Polar-orbiting Partnership (Suomi NPP), 1=NOAA-20 (designated JPSS-1 prior to launch) for VIIRS and A= Aqua and T=Terra for MODIS
- **instrument**: MODIS or VIIRS
- **confidence**: This value is based on a collection of intermediate algorithm quantities used in the detection process. It is intended to help users gauge the quality of individual hotspot/fire pixels. Confidence values are set to low, nominal and high or between 0-100%.

1. K-NEIGHBORING NEIGHBORS (KNN)

It is a Classification method most commonly used to implement learning based on the k nearest neighbors, where k is an integer value specified by the user and hence, differs with each dataset as its value is highly data-dependent. Therefore, if we choose a higher k, it makes classification boundaries less distinct. It usually uses 'uniform' weights for each query point but if the data is not uniformly sampled then we can give higher weights to the neighbors near k and the weights are proportional to the inverse of distance from the query point.

Since our data is quite uniform, we will use 'uniform' weights for the KNN Classification. To get accurate results, we will train the dataset first by randomly dividing it into 80-20 and then testing 20% of it out. We will also use accuracy score and precision score to ensure the results derived are highly accurate.

2. LOGISTIC REGRESSION

It is used to derive a model that estimates probability of an observation x belonging to class c from the training data. The odds of success is defined as $\mathcal{O}(x) = p/(1-p)$ and its logarithm is known as the log-odds where $p = P[C = 1|x]$ which is the probability of the true class label being C = 1 given observations x.

Since, it can only be used on continuous valued features - we will predict the probability of bushfire by using the numerical features such as 'scan' and 'track', their values being converted to binary - 0 if less than (or equal to) 0.5 or 1 if greater than 0.5, and finding their regression relation with column 'confidence'. The quantiles for the feature - 'confidence' will be calculated and then accordingly the values will be converted to 0 if less than 58%, 1 if between 58%-80% meaning no to low optimal fire conditions, 2 for values between 80%-98% referring to nominal conditions for fire to occur and 3 for 98%-100% highly indicating that the fire will occur. Hopefully, if its accuracy is quite good (> 80%) then it will greatly benefit the environment we are living in, as we can take necessary precautions so

that bushfires can be prevented from being turned into forest fires and the loss of local flora and fauna can also be avoided.

3. MULTI-LAYER PERCEPTRON

It is a supervised learning method used to determine a function by training on a dataset, we can have any number of input features, hidden layers and any number of neurons per hidden layer. The function f can be linear or non-linear - $f: \mathbb{R}^m \rightarrow \mathbb{R}^o$, where m is the number of dimensions for input and o is the number of dimensions for output. Its ability to learn models in real-time will help in analysis of satellite imagery which is an added advantage. This will especially help in determining a function or relation between brightness temperature of the pixel recorded by the satellite and the probability of fire. Furthermore, it can even help us to detect which areas are more susceptible to bushfire or the areas where the fire often originates. But Multi-Layer Perceptron requires tuning a number of hyperparameters such as the number of hidden neurons, layers and iterations and is sensitive to feature scaling which will make it tedious to determine the relationship between them and affect the certainty of the function obtained.

4. NEURAL NETWORKS

The data can be analyzed using neural networks to find intricate patterns and connections between the input features and the target variable. Neural networks' capacity to process huge amounts of data is one of their advantages. Data from satellite imagery can have thousands or even millions of features, making it highly high-dimensional. When processing this data, neural networks can efficiently spot trends and connections that other machine learning algorithms might find challenging to spot.

We must use the preprocessed data and the chosen characteristics to train the neural network model. This entails randomly dividing the data into training and validation groups, feeding the data into the neural network, adjusting the weights and biases, and repeating the procedure until the model reaches an acceptable level of accuracy.

It is crucial to keep in mind that neural networks need a lot of high-quality training material to function properly.

5. UNSUPERVISED LEARNING

Unsupervised learning can be especially helpful in the case of bushfire prediction using satellite images because it can find patterns and anomalies in the data that may indicate the probability or severity of a bushfire.

Using variables like brightness, temperature, and location, clustering can be used to put similar satellite images into groups for bushfire prediction. This can be used to pinpoint areas where bushfires are more likely to commence or where one may already be burning.

Anomaly detection, which entails locating data points that greatly deviate from the norm, is another unsupervised learning technique that can be helpful for bushfire prediction. Anomaly detection can be used to locate areas of satellite imagery where there is an unusual quantity of brightness or other anomalous patterns that might be signs of a bushfire.

Unsupervised learning can also be used with a variety of data sets and can help find patterns that more conventional supervised learning techniques might overlook because it does not require labelled training data.

FINAL MODEL

The analysis of different machine learning methods applied to the dataset will determine the final bushfire prediction model. Although reducing the false positive rate is crucial, in this instance, the emphasis should be on maintaining a low false negative rate. This is due to the possibility that false negatives could prevent the detection and response to an actual fire, which could cause substantial ecological harm and the extinction of wildlife. To accurately spot bushfires while avoiding false negatives, this necessitates a careful assessment of the model's performance, strengths, and limitations. Therefore, the model with the highest success rate and the lowest false negative rate will be the one that best forecasts and prevents bushfires.

REFERENCES

[1] "ENGG2112 LECTURE SLIDES."

https://canvas.sydney.edu.au/courses/48636/files/30262966?module_item_id=1929221

(accessed Mar. 30, 2023).

[2] Wikipedia Contributors, "2019–20 Australian bushfire season," Wikipedia, Jan. 12, 2020.

https://en.wikipedia.org/wiki/2019%E2%80%9220_Australian_bushfire_season (accessed Mar.

31, 2023).

[3] Bureau of Meteorology, "Bushfire weather," Bom.gov.au, 2017.

<http://www.bom.gov.au/weather-services/fire-weather-centre/bushfire-weather/index.shtml>

(accessed Apr. 02, 2023).

[4] scikit-learn, "scikit-learn: machine learning in Python," Scikit-learn.org, 2019.

<https://scikit-learn.org/stable/> (accessed Mar. 28, 2023).

[5] Scikit-Learn, "User guide: contents — scikit-learn 0.22.1 documentation," Scikit-learn.org, 2019.

https://scikit-learn.org/stable/user_guide.html (accessed Mar. 28, 2023).

DATASET

[1] "Australian Bush fire satellite data (NASA)," *www.kaggle.com*, 2019.

https://www.kaggle.com/datasets/nagarajbhat/australian-bush-fire-satellite-data-nasa?select=fire_archive_M6_101673.csv

(accessed Apr. 02, 2023).